

- Numerical integration -

Exercise 1: Newton-Cotes formulas

We define the function $f(x) = \frac{1}{x}$.

- (1) Calculate the polynomial $P_1(x)$ of degree less than or equal to 1, which approximates the function f in the sense of continuous least squares over $[\frac{1}{2}, 1]$.
- (2) Calculate $\int_{\frac{1}{2}}^1 \frac{1}{x} dx$ with 2 methods.

Exercise 2: About integration

- (1) Determine, using the Newton basis, the Lagrange interpolation polynomial P_2 associated with the nodes $x_0 = 1$, $x_1 = \frac{3}{2}$, and $x_2 = 2$ of the function $f(x) = \frac{1}{x}$.
- (2) Using the interpolation error formula, show that:
$$\forall x \in [1, 2], \quad |f(x) - P_2(x)| \leq (x-1)(2-x) \left| x - \frac{3}{2} \right|.$$
- (3) Give the linear system satisfied by $(\lambda_0, \lambda_1, \lambda_2)$ so that the method J_2 is exact for all polynomials of degree less than or equal to 2.
- (4) Deduce the values of λ_0 , λ_1 , and λ_2 .
- (5) What is the degree of exactness of the method?
- (6) Deduce from this study an approximate value of $\ln(2)$. What could be done to obtain an approximation as accurate as desired?

Exercise 3: Approximating a Marginal Probability in a Bayesian Network

We consider the marginal probability:

$$P(E = e) = \int_0^1 P(E = e \mid X = x) \cdot P(X = x) dx$$

with the following definitions:

$$P(E = e \mid X = x) = \sin(\pi x), \quad P(X = x) = x.$$

- (a) Approximate the integral using the trapezoidal rule with $n = 4$ subintervals.
- (b) Approximate the same integral using Simpson's rule with $n = 4$.
- (c) Compare the two results with the exact value computed symbolically.

Exercise 4: Generalized Trapezoidal/Simpson methods

Let the integral $I = \int_0^\pi \sin x \, dx$.

- (1) Calculate the exact value of I .
- (2) Using the generalized trapezoidal method and the generalized Simpson's method for $h = \frac{\pi}{4}$.
 - (a) Calculate I ;
 - (b) Estimate the error;
 - (c) Evaluate the error.
- (3) Provide the value of step h and the number of subdivisions of the interval $[0, \pi]$ so that the error obtained by the generalized trapezoidal method (resp. Simpson's) is less than 5×10^{-4} .

Exercise 5: Estimating Area of a Robot's Sensor Field

An autonomous robot scans its environment every 2 meters and records the detection width (in meters) of its sensor. The recorded values at regular intervals are:

$$f(x_0) = 1.8, \quad f(x_1) = 2.4, \quad f(x_2) = 2.6, \quad f(x_3) = 2.3, \quad f(x_4) = 2.0, \quad f(x_5) = 1.7, \quad f(x_6) = 1.5$$

Assuming the measurements were taken at equal intervals of $h = 2$ meters, use Simpson's Rule to estimate the area scanned by the robot.

Exercise 6: A problem

The widths (in meters) of a kidney-shaped swimming pool were measured at 2-meter intervals, as indicated in the figure. Use Simpson's Rule to estimate the area of the pool.

